

All these words! No experience is more familiar to us than this ceaseless barrage of words popping up in our mind extremely efficiently and without ever being invited. But where do these words come from, and what kind of invisible mechanism makes them bubble up? What is going on when one merely thinks silently to oneself, “a mother and her daughter”?

It All Starts with Single-member Categories

To be able to attach the label “mother” to some entity without thinking about it, one has to be intimately familiar with the concept *mother*, which is denoted by the word. For most of us, this intimacy with the concept goes all the way back to our earliest childhood, to our first encounters with the notion. For one-year-old Tim, the core of the concept is clearly his own mother — a person who is much bigger than he is, who feeds him, comforts him when he cries, sings him lullabies, picks him up, plays with him in the park, and so forth. Once this first mental category bearing the name “Mommy” has a toehold, Tim will be able to see that in the world around him there are similar phenomena, or as we prefer to put it, analogous phenomena.

We take a momentary break here to explain a typographical convention of our book. When speaking about a word, we will put it in quotation marks (“table”), whereas when speaking about a concept, we will use italics (*table*). This is an important distinction, because whereas a word is a sequence of sounds, a set of printed letters, or a chunk of silent inner language, a concept is an abstract pattern in the brain that stands for some regular, recurrent aspect of the world, and to which any number of different words — for instance, its names in English, French, and so forth, or sometimes no word at all — can be attached. Words and concepts are different things. Although the distinction between them is crucial and often very clear, there will unavoidably be cases in our text where it will be ambiguous and blurry, and in such cases, we’ll make a choice between italics and quotation marks that might seem a bit arbitrary. Another source of ambiguity is the fact that here and there we’ll use italics for emphasis, just as we’ll use quotation marks to suggest a sense of doubt or approximation (which could sometimes be conveyed equally well by the word “so-called”), and of course we will use quotation marks when we are making a quotation. Alas, the world is simply filled with traps, but we hope that the ambiguities are more theoretical than actual. And with that said, we return to our main story.

One day in the park, Tim, aged eighteen months, sees a tot playing in the sandbox and then notices a grown-up near her who is taking care of her. In a flash, Tim makes a little mental leap and thinks to himself more or less the following (although it’s far from being fully verbalized): “That person is taking care of *her* just like Mommy takes care of *me*.” That key moment marks the birth of the concept *mommy* with a small “m”. The lowercase letter is because there are two members of this new category now (and of course using uppercase and lowercase letters is just *our* way of hinting at what’s going on in Tim’s head, not *his* way). From this point on, it won’t take Tim long to notice yet other instances of this concept.

At the outset, Tim's concept of *mommy* still floats between singular and plural, and the analogies in his head will be quite concrete, a comparison always being made to the *first* mommy, which is to say, with Mommy (the one with the capital "M"), but as new instances of the concept *mommy* are superimposed and start to blur in his memory, the mental mapping that Tim will automatically carry out, each time he spots a new grown-up in the park, will start to be made not onto Mommy, but onto the nascent and growing concept of *mommy* — that is, onto a generalized, stereotyped, and even slightly abstract situation, centered on a generic grown-up (*i.e.*, stripped of specific details) and involving a generic child who is near the grown-up and whom the grown-up talks to, smiles at, picks up, comforts, watches out for, and so on.

It's not our goal here to lay out a definitive theory of the growth of the specific concept *mommy*, as our purpose is more general than that. What we are proposing is that the birth of *any* concept takes place more or less as described above. At the outset, there is a concrete situation with concrete components, and thus it is perceived as something unique and cleanly separable from the rest of the world. After a while, though — perhaps a day later, perhaps a year — one runs into another situation that one finds to be similar, and a link is made. From that moment onward, the mental representations of the two situations begin to be connected up, to be blurred together, thus giving rise to a new mental structure that, although it is less specific than either of its two sources (*i.e.*, less detailed), is not fundamentally different from them.

And so the primordial concept *Mommy* and the slightly more sophisticated concept *mommy* act in very similar ways. In particular, both of them are easily mapped onto newly encountered situations "out there", which leads both of them to extending themselves outwards — a snowball effect that will continue all throughout life. It's this idea of concepts extending themselves forever through a long series of spontaneous analogies that we wish to spell out more carefully in the next few sections.

Passing from *Mommy* to *mommy* and then to *mother*

One day, Tim, who sadly has never met his father, is playing in the park, and he runs into a little girl accompanied by a grown-up who is encouraging the girl to play with the other children. He thinks to himself that this grown-up is the *mommy* of the little girl. That is, Tim's mind makes a link between what he's observing and his new concept of *mommy*. This is an act of categorization. Perhaps the new person is not actually the child's mother but the child's father, or perhaps it's her grandmother, or even her older brother or sister, but even so, that doesn't make Tim's mapping of this new person onto the category *mommy* irrational, because his notion of *Mommy/mommy* is wider than ours is (not richer, of course, but less discriminating, due to his lack of experience). This simple analogy Tim has made is flawless; it's just that he hasn't taken into account certain details that an adult would have used. If Sue, his mother, explains to him that this person isn't the little girl's *mommy* but her *daddy*, then Tim may well modify his concept of *mommy*, thereby coming into closer alignment with the people around him.

Gradually, as Tim uses the word “mommy” more and more, his initial image — that of his own mother — will start to recede from view, like a root being grown over ever more as time passes. He will overlay his earliest image with traits of other people whom he assigns to this mental category, and the vivid and unique features of his own Mommy will become harder and harder to find in it. Nonetheless, even when Tim is himself a grown-up, there will remain in his concept of *mommy* some residual traces of his primordial concept *Mommy*.

One day, a friendly woman who’s come all the way from her home in Canada turns up and treats Tim very sweetly. He hears the word “mommy” used several times to refer to this grown-up, and so for a while he concludes that maybe he has more than one mommy. For Tim this is conceivable, since he has not yet built up a set of expectations that would rule this possibility out. Sometimes his “second mommy” takes him to the park and she, too, chats with the other mommies. But after a week or so, Tim’s second mommy vanishes, which quite understandably saddens him. The next day, one of the mommies in the park asks Tim, “Did your grandmother go back home?” Tim doesn’t answer, because he doesn’t yet know the concept of *grandmother*. So she reformulates her question: “Where’s your mommy’s mom today, Tim?” But this question makes even less sense to Tim. He knows perfectly well that *he’s* the one who has the mommy (he even had two of them in the past few days!), and so his mommy (that is, the remaining one) can’t have a mommy. After all, it’s *children* who have mommies (and sometimes also daddies) whose purpose is to be sweet to them, to watch over them, and to help them, and Tim knows that his mommy *isn’t* a child, and so she doesn’t have a mommy. That’s obvious! The woman doesn’t push her strange question, and Tim goes back to his playing.

And time passes. A few months later, Tim starts to realize that grown-ups are sometimes accompanied by other grown-ups that they refer to as their “mother”. Suddenly everything starts to be clear... What children have are *mommies*, and what grown-ups have are *mothers*. That makes sense! And into the bargain, there’s even an analogical bond between *mommy* and *mother*. Of course Tim isn’t aware of having made an analogy — neither this concept nor the word for it will be known to him for another ten or more years! — but he has nonetheless made one. And as is often the case with analogies, this one helps clarify things for Tim but it also misleads him a little.

We now will skip over the details, simply adding that the two concepts of *mommy* and *mother* gradually merge to create a more complex concept at whose core there is the primordial concept of *Mommy*. This doesn’t mean that the primordial image of Sue springs to Tim’s mind every time that he hears the word “mother” or even the word “mommy”, but merely that the invisible roots are structured in that manner.

As any concept grows in generality, it also becomes more discriminating, which means that at some point it’s perfectly possible that some early members of the category might be demoted from membership while new members are being welcomed on board. Thus the dad at the park whom Tim had first taken for a *mommy* is stripped of the label, and although Tim’s grandmother stays on as a member of the category *mother*, she winds up in a less central zone than the *mommy* zone, which is reserved for the

mothers of small children. And of course as time goes by, Tim will come to understand that his grandmother herself was once a member of the category *mommy* (just as his own Mommy was once a member of the category *small child*), but at present all of that is well beyond his grasp.

The Cloud of Concepts of *Mother*

One might think that the concept of *mother* is very precise — perhaps as precise as that of *prime number*. That would imply that to every question of the form “Is X a mother or not?”, there would always be a correct, objective, black-and-white answer. But let’s consider this for a moment. If a little girl is playing with two dolls, one bigger and one smaller, and she says that the big one is the small one’s mother, is this an example of motherhood? Does the large doll belong to the category *mother*? Or contrariwise, could one state without risk of contradiction that she does *not* belong to that category?

And if we read a certain book in which a certain Sue is described as the mother of a certain Tim, then does this Sue, who is never anything but a made-up character in a book, truly belong to the category *mother*? Does it make any difference that Sue was modeled on a real person, and Tim on her son? Is Sue more of a mother than the doll is? What indeed is Sue? If in the book it states that she is 34 years old, that she has light brown hair, that she weighs 120 pounds, that she is five feet five inches tall, and that she’s the mom of a small boy, does that mean that Sue has a body and once gave birth? A doll, at least, is a physical object, but what is Sue, when you come down to it? An abstract thought triggered by some words on a page, by some black marks on a white background. Does this thought even deserve the pronoun “she”?

When Tim gets to be six, if someone tells him that Lassie is Spot’s mother, he certainly won’t protest, but if he were told that the queen bee is the mother of all the bees in her hive, it’s less clear what he would say, and in any case some mental effort would be needed before he could absorb this idea. And if he were told that a drop of water that he just watched dividing into two drops is the mother of the two new drops, he would almost surely find this suggestion very surprising. Everyone knows phrases that use the word “mother” in ways that go far beyond the senses that apply to Lassie, the queen bee, or even the splitting drop of water — for instance, “my motherland”, “a mother cell”, “the mother lode”, “Mother Earth”, “Greece is the mother of democracy”, and “Necessity is the mother of invention”. Are these true instances of the concept of *mother*, genuine cases of maternity? What is the proper way to understand such usages of the word?

Some readers may feel inclined to say that these are all “metaphorical mothers”, and indeed, such a viewpoint is not without merit, but we have to point out that there is no sharp boundary that separates “true” mothers from those that are metaphorical, for categories in general don’t have sharp boundaries; most of the time, metaphorical and literal meanings overlap so greatly that when one tries to draw a clear boundary, one discovers that things only get blurrier and blurrier.